

Intelligent Agents

Introduction (5 Minutes)

Imagine you are driving a car. How do you know when to brake?

- I see another vehicle.
- I hear a horn.
- I observe a traffic signal.

Who actually decides to press the brake?

The driver.

A driver continuously:

- Observes the environment.
- Thinks about what is happening.
- Decides what action to take.
- Performs that action.

An **Artificial Intelligence Agent** behaves in the same way.

What is an Agent?

An **Agent** is anything that can:

- Observe its environment.
- Make decisions.
- Perform actions.

Definition (Russell & Norvig)

An agent is anything that perceives its environment through sensors and acts upon that environment through actuators.

Examples of Agents

Agent	Sensors	Actuators
Human	Eyes, ears, nose, skin	Hands, legs, mouth
Robot	Camera, microphone, ultrasonic sensor	Wheels, robotic arms
Self-driving car	Cameras, LiDAR, Radar, GPS	Steering, brakes, accelerator
Smartphone	Camera, GPS, microphone	Display, speaker, vibration

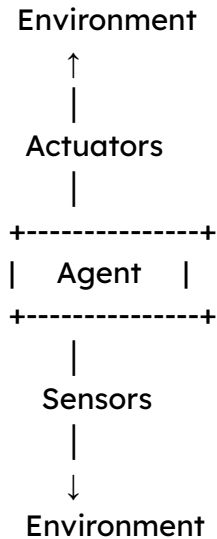
Smart Vacuum
Cleaner

Dirt sensor, bump sensor

Wheels, suction motor

Basic Agent Model

Draw the following diagram on the board:



Explain:

- Sensors collect information.
- Agent processes information.
- Actuators perform actions.
- Environment changes.

This process repeats continuously.

What are Sensors?

Sensors collect information from the environment.

Examples:

Human:

- Eyes
- Ears
- Nose

Robot:

- Camera
- Temperature sensor
- GPS
- Microphone
- Infrared sensor

Without sensors,
the agent cannot know what is happening.

What are Actuators?

Actuators perform actions.

Examples:

Human:

- Hands
- Legs
- Mouth

Robot:

- Wheels
- Robotic arm
- Speaker
- Display

Without actuators,
the agent cannot interact with the environment.

Example: Self-driving Car

Sensors:

- Cameras
- GPS
- Radar
- LiDAR

Agent:

- AI software

Actuators:

- Steering
- Accelerator
- Brake

Environment:

- Roads
- Vehicles
- Pedestrians
- Traffic signals

Agent Function

This is an important exam topic.

Definition:

An **Agent Function** specifies what action should be taken for every possible percept sequence.

In simple words:

It tells the agent:

"If this situation occurs, perform this action."

Example:

Perception	Action
Red Light	Stop
Green Light	Move
Obstacle	Turn Right
Person Crossing	Brake

The agent function is **abstract**—it describes the desired mapping from perceptions to actions.

Percepts and Percept Sequence

Percept

A **Percept** is the information an agent receives from its sensors at a given instant.

Example:

A camera detects:

- Red traffic light

This observation is one percept.

Percept Sequence

A **Percept Sequence** is the complete history of all percepts received.

Example:

1. Road is empty
2. Green signal
3. Pedestrian appears
4. Red signal

The sequence helps the agent make better decisions.

Agent Program

The **Agent Program** is the actual software that implements the agent function.

Think of it as the computer code that runs on the hardware.

Example:

Camera captures image → program detects traffic light → program decides to stop
→ brake is applied.

Agent Architecture

Agent Architecture is the physical platform on which the program runs.

Examples:

- Computer
- Robot hardware
- Mobile phone
- Drone
- Raspberry Pi

Relationship

Write this on the board:

Agent = Architecture + Program

Explain:

- Architecture = Hardware
- Program = Software
- Together they create the intelligent agent.

Example:

Hardware:

- Robot

Software:

- Navigation algorithm

Together:

- Robot that can move intelligently.

Characteristics of Intelligent Agents

An intelligent agent should have the following properties:

1. Autonomy

Works without continuous human intervention.

Example:

Robot vacuum cleaner.

2. Reactivity

Responds to environmental changes.

Example:

Automatic doors.

3. Proactiveness

Acts to achieve goals instead of only reacting.

Example:

Google Maps suggesting a faster route before you encounter traffic.

4. Learning Ability

Improves performance based on experience.

Example:

Netflix recommendations become more personalized over time.

5. Rationality

Chooses the action expected to achieve the best outcome based on available information.

Example:

A chess program selects the strongest move it can evaluate.

Types of Intelligent Agents

1. Simple Reflex Agent

Uses current percept only.

Rule:

IF condition THEN action

Example:

IF room is dirty

THEN clean room

Advantages:

- Simple
- Fast

Disadvantages:

- Cannot remember previous events.

2. Model-Based Reflex Agent

Maintains an internal model (memory) of the world.

Example:

Robot remembers:

"I cleaned Room A already."

Better decisions because it uses both current percepts and stored information.

3. Goal-Based Agent

Has a specific goal.

Example:

GPS Navigation

Goal:

Reach destination.

The agent evaluates different routes to achieve that goal.

4. Utility-Based Agent

Chooses the action with the highest utility (preference or usefulness).

Example:

Navigation system chooses:

- Shortest route
- Fastest route
- Safest route

based on what provides the greatest benefit.

5. Learning Agent

Improves through experience.

Example:

ChatGPT

Learns from large datasets during training and improves through updated models and feedback.

Other examples:

- Spam filters
- Recommendation systems

Comparison of Agent Types

Agent Type	Memory	Goals	Learns
Simple Reflex	No	No	No
Model-Based	Yes	No	No
Goal-Based	Yes	Yes	No
Utility-Based	Yes	Yes	No
Learning Agent	Yes	Yes	Yes

Real-Life Examples

ATM Machine

Sensors:

- Keypad
- Card reader

Actuators:

- Cash dispenser

- Display

Google Maps

Sensors:

- GPS
- Traffic data

Actuators:

- Screen
- Voice guidance

Alexa

Sensors:

- Microphone

Actuators:

- Speaker

Self-driving Car

Sensors:

- Cameras
- Radar
- LiDAR

Actuators:

- Brake
- Steering
- Accelerator

Summary

Students should remember:

- An agent perceives the environment through **sensors** and acts through **actuators**.
- A **percept** is a single input from the environment; a **percept sequence** is the history of inputs.
- An **agent function** maps percept sequences to actions.
- An **agent program** implements the agent function on an **agent architecture**.
- **Agent = Architecture + Program.**
- Intelligent agents are autonomous, reactive, proactive, capable of learning, and strive to act rationally.
- The five main agent types are **Simple Reflex, Model-Based Reflex, Goal-Based, Utility-Based, and Learning Agents.**